
Web Fundamentals & HTML Basics

Beginner-Friendly Notes

A complete guide for students new to web development

Topics Covered

HTML Document Structure · Tags & Attributes
Text Formatting · Links · Images · Lists
Tables · Forms · Semantic HTML

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1 What is HTML?

HTML stands for **HyperText Markup Language**. It is the standard language used to create and design web pages. Every website you visit on the internet is built using HTML at its core.

Think of HTML as the *skeleton* of a web page — it gives structure to content like headings, paragraphs, images, and links. HTML is **not** a programming language; it is a **markup language**, which means it uses special tags to annotate (mark up) text so a web browser knows how to display it.

1.1 Key Facts

- HTML files are plain text files saved with a `.html` or `.htm` extension.
- Web browsers (Chrome, Firefox, Edge, Safari) read HTML and render it visually.
- HTML works together with **CSS** (for styling) and **JavaScript** (for behaviour).
- The current standard is **HTML5**, introduced in 2014.

Tip

You do not need any special software to write HTML. A simple text editor like **Notepad** (Windows), **TextEdit** (Mac), or **VS Code** is enough to get started.

1.2 How Does HTML Work?

HTML uses **tags** to mark up content. Tags are keywords enclosed in angle brackets. Most tags come in pairs — an **opening tag** and a **closing tag**:

```
1 <tagname> Content goes here </tagname>
```

Listing 1: Basic HTML tag structure

The closing tag has a forward slash (/) before the tag name. Some tags are **self-closing** (also called void elements) and do not need a closing tag, for example `<br>` and `<img>`.

2 Structure of an HTML Document

Every HTML page follows a standard structure. Below is the minimal template you should start with for every HTML file:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-
6       scale=1.0">
7     <title>My First Web Page</title>
8   </head>
9   <body>
10    <h1>Hello, World!</h1>
```

```
10 <p>Welcome to my first web page.</p>
11 </body>
12 </html>
```

Listing 2: Basic HTML5 document template

Let us examine each part of this template in detail.

2.1 <!DOCTYPE html>

This is the very first line of every HTML5 document. It is a **declaration** (not a tag) that tells the browser which version of HTML is being used.

- <!DOCTYPE html> tells the browser: “This is an HTML5 document.”
- It must appear before the <html> tag.
- It is not case-sensitive, but writing it in uppercase is conventional.

Note

Without <!DOCTYPE html>, the browser may switch to *quirks mode* and render the page incorrectly.

2.2 <html>

The <html> tag is the **root element** of the page. All other elements are nested inside it. The **lang** attribute specifies the language of the page content (e.g., “en” for English).

```
1 <html lang="en">
2   <!-- All page content goes here -->
3 </html>
```

2.3 <head>

The <head> element contains **metadata** — information *about* the page that is not displayed to the user. Typical content includes:

- <title> — the page title shown on the browser tab.
- <meta> — character encoding, viewport settings, description.
- <link> — links to external CSS stylesheets.
- <script> — links to JavaScript files.

2.4 <title>

The <title> tag sets the text that appears in the **browser tab** and in search engine results.

```
1 <head>
2   <title>My Awesome Website</title>
3 </head>
```

Tip

Choose a meaningful, descriptive title — it helps with Search Engine Optimisation (SEO) and tells users what the page is about.

2.5 <body>

The `<body>` element contains **all the visible content** of the web page — everything the user actually sees in the browser window: headings, paragraphs, images, links, tables, and so on.

```
1 <body>
2   <h1>Welcome!</h1>
3   <p>This text is visible in the browser.</p>
4 </body>
```

3 Headings (h1 to h6)

HTML provides **six levels of headings**, from `<h1>` (largest and most important) to `<h6>` (smallest and least important). Headings help organise content and make pages easier to read.

```
1 <h1>Heading Level 1 - Main Title</h1>
2 <h2>Heading Level 2 - Section Title</h2>
3 <h3>Heading Level 3 - Subsection</h3>
4 <h4>Heading Level 4</h4>
5 <h5>Heading Level 5</h5>
6 <h6>Heading Level 6 - Smallest Heading</h6>
```

Listing 3: All six heading levels

Remember

Use only **one** `<h1>` per page — it is the main title. Headings should follow a logical hierarchy; do not skip levels (e.g., go from `<h2>` straight to `<h4>`).

Note

Headings are also important for **accessibility** — screen readers use them to help visually impaired users navigate the page.

4 Paragraphs, Line Breaks, and Horizontal Rules

4.1 Paragraph Tag <p>

The `<p>` tag defines a **paragraph** of text. Browsers automatically add some vertical space (margin) before and after each paragraph.

```
1 <p>This is the first paragraph. It contains some text.</p>
2 <p>This is the second paragraph. Browsers add space between
   them.</p>
```

```
3 <p>Each paragraph starts on a new line automatically.</p>
```

Listing 4: Using paragraph tags

Note

Extra spaces and blank lines inside HTML code are **ignored** by browsers. Use tags like `<p>` and `<br>` to control spacing.

4.2 Line Break `<br>`

The `<br>` tag inserts a **single line break** without starting a new paragraph. It is a **self-closing** (void) element — it has no closing tag.

```
1 <p>
2   123 Main Street<br>
3   Springfield<br>
4   United States
5 </p>
```

Listing 5: Using line breaks

Tip

Use `<br>` sparingly — for addresses and poems, not to create extra spacing. For spacing, use CSS margin/padding instead.

4.3 Horizontal Rule `<hr>`

The `<hr>` tag draws a **horizontal line** across the page. It is used to separate sections of content visually. Like `<br>`, it is a self-closing element.

```
1 <h2>Chapter 1</h2>
2 <p>Content of chapter one goes here.</p>
3
4 <hr>
5
6 <h2>Chapter 2</h2>
7 <p>Content of chapter two goes here.</p>
```

Listing 6: Using horizontal rules

5 Text Formatting Tags

HTML offers several tags to format and emphasise text. Some tags are **presentational** (visual only) while others carry **semantic meaning** (they communicate importance to browsers and screen readers).

```
1 <p>This text is <b>bold</b> using the b tag.</p>
2 <p>This text is <strong>strongly important</strong> using
   strong.</p>
```

```
3 <p>This text is <i>italic</i> using the i tag.</p>
4 <p>This text is <em>emphasised</em> using em.</p>
5 <p>This text is <u>underlined</u> using the u tag.</p>
6 <p>This text is <mark>highlighted</mark> using mark.</p>
7 <p>This text is <small>smaller</small> using the small tag.</p>
```

Listing 7: Text formatting tags in action

5.1 Bold: `<b>` and `<strong>`

- `<b>` — makes text **visually bold** with no extra semantic meaning.
- `<strong>` — makes text bold **and** signals that it is of high importance. Screen readers may read it with added emphasis.

Prefer `<strong>` when the content is genuinely important.

5.2 Italic: `<i>` and `<em>`

- `<i>` — renders text in *italic style* (used for technical terms, foreign words, or titles).
- `<em>` — italicises text **and** adds *semantic emphasis*, which screen readers may stress when reading aloud.

5.3 Underline: `<u>`

The `<u>` tag underlines text. Use it carefully — underlined text is often mistaken for a hyperlink.

5.4 Highlight: `<mark>`

The `<mark>` tag highlights text with a yellow background by default, similar to a highlighter pen.

```
1 <p>Remember: <mark>HTML5 is the current standard.</mark></p>
```

5.5 Small Text: `<small>`

The `<small>` tag renders text in a smaller font size. It is commonly used for fine print, copyright notices, and disclaimers.

```
1 <p>Buy now! <small>(Terms and conditions apply.)</small></p>
```

6 Links `<a>`

Links (hyperlinks) allow users to navigate between pages. The `<a>` (anchor) tag creates a clickable link. The most important attribute is `href` (Hypertext Reference), which specifies the URL destination.

```
1 <!-- Link to an external website -->
2 <a href="https://www.example.com">Visit Example Website</a>
3
4 <!-- Link that opens in a new browser tab -->
5 <a href="https://www.example.com" target="_blank">Open in New
  Tab</a>
6
7 <!-- Link to another page in the same site -->
8 <a href="about.html">About Us</a>
9
10 <!-- Link to a section on the same page (using id) -->
11 <a href="#section2">Jump to Section 2</a>
12
13 <!-- Email link -->
14 <a href="mailto:hello@example.com">Send us an Email</a>
```

Listing 8: Creating links

6.1 Important Attributes of <a>

Attribute	Description
href	The URL or path the link points to
target	Where to open the link ("blank" = new tab, "self" = same tab)
title	Tooltip text shown when hovering over the link
rel	Relationship between the current and linked page (e.g., "noopener")

Remember

When using `target="_blank"`, always add `rel="noopener noreferrer"` for security reasons. This prevents the new tab from accessing your page.

7 Images

The `<img>` tag embeds an image in a web page. It is a **self-closing** (void) element — it does not have a closing tag.

```
1 <!-- Basic image -->
2 
3
4 <!-- Image from the web (URL) -->
5 
6
7 <!-- Image with specified width and height -->
8 
```

Listing 9: Embedding images

7.1 Essential Attributes

Attribute	Description
<code>src</code>	Path or URL to the image file
<code>alt</code>	Alternative text description (shown if image cannot load)
<code>width</code>	Width of the image in pixels (or percentage)
<code>height</code>	Height of the image in pixels (or percentage)
<code>title</code>	Tooltip text displayed on hover

Note

The `alt` attribute is **required** for accessibility. Screen readers read the alt text aloud to visually impaired users. It also appears when the image fails to load.

Tip

Supported image formats include **JPEG** (.jpg), **PNG** (.png), **GIF** (.gif), **SVG** (.svg), and **WebP** (.webp).

8 Lists

HTML provides three types of lists to organise content: **unordered**, **ordered**, and **description** lists.

8.1 Unordered List

An unordered list displays items with **bullet points**. Each item uses the `<li>` (list item) tag.

```
1 <h3>My Favourite Fruits</h3>
2 <ul>
3   <li>Apple</li>
4   <li>Mango</li>
5   <li>Banana</li>
6   <li>Strawberry</li>
7 </ul>
```

Listing 10: Unordered list example

8.2 Ordered List

An ordered list displays items with **numbers** (or letters). Use it when the sequence matters, such as steps in a process.

```
1 <h3>How to Make Tea</h3>
2 <ol>
3   <li>Boil water in a kettle.</li>
4   <li>Place a tea bag in a cup.</li>
5   <li>Pour the boiling water into the cup.</li>
6   <li>Wait 3 minutes, then remove the tea bag.</li>
```

```

7   <li>Add milk and sugar if desired.</li>
8 </ol>

```

Listing 11: Ordered list example

8.3 Nested Lists

Lists can be **nested** inside one another to create sub-items:

```

1 <ul>
2   <li>Fruits
3     <ul>
4       <li>Apple</li>
5       <li>Mango</li>
6     </ul>
7   </li>
8   <li>Vegetables
9     <ul>
10      <li>Carrot</li>
11      <li>Spinach</li>
12    </ul>
13  </li>
14 </ul>

```

Listing 12: Nested list example

Tip

Use `<ul>` for items without a specific order (shopping lists, features). Use `<ol>` for sequential steps (recipes, instructions, rankings).

9 Tables

HTML tables are used to display **tabular data** — information organised in rows and columns (like a spreadsheet). The key tags are:

Tag	Purpose
<code><table></code>	Defines the entire table
<code><tr></code>	Table Row — creates a horizontal row
<code><th></code>	Table Header — bold, centred header cell
<code><td></code>	Table Data — regular data cell
<code><thead></code>	Groups the header rows (optional, recommended)
<code><tbody></code>	Groups the body rows (optional, recommended)
<code><caption></code>	Adds a title above the table

```

1 <table border="1">
2   <caption>Student Marks</caption>
3   <thead>
4     <tr>
5       <th>Name</th>

```

```
6      <th>Subject</th>
7      <th>Marks</th>
8      <th>Grade</th>
9  </tr>
10 </thead>
11 <tbody>
12   <tr>
13     <td>Alice</td>
14     <td>Mathematics</td>
15     <td>92</td>
16     <td>A</td>
17   </tr>
18   <tr>
19     <td>Bob</td>
20     <td>Science</td>
21     <td>85</td>
22     <td>B</td>
23   </tr>
24   <tr>
25     <td>Carol</td>
26     <td>English</td>
27     <td>78</td>
28     <td>C</td>
29   </tr>
30 </tbody>
31 </table>
```

Listing 13: A complete HTML table

9.1 Spanning Cells

Cells can span multiple rows or columns using:

- `colspan="n"` — makes a cell span n columns.
- `rowspan="n"` — makes a cell span n rows.

```
1 <table border="1">
2   <tr>
3     <th colspan="2">Name and Age</th>
4     <th>City</th>
5   </tr>
6   <tr>
7     <td>Alice</td>
8     <td>22</td>
9     <td rowspan="2">New York</td>
10  </tr>
11  <tr>
12    <td>Bob</td>
13    <td>25</td>
14  </tr>
15 </table>
```

Listing 14: Table with colspan and rowspan

Remember

Tables should be used for **tabular data only** — not for page layout. Use CSS Flexbox or Grid for page layouts.

10 Forms

HTML forms allow users to **input data** and send it to a server (e.g., login pages, registration forms, search boxes, contact forms). The `<form>` element is the container for all form elements.

```

1  <form action="/submit" method="post">
2
3      <!-- Text input -->
4      <label for="name">Full Name:</label>
5      <input type="text" id="name" name="name" placeholder="Enter
        your name">
6
7      <!-- Email input -->
8      <label for="email">Email Address:</label>
9      <input type="email" id="email" name="email" placeholder="
        you@example.com">
10
11     <!-- Password input -->
12     <label for="pwd">Password:</label>
13     <input type="password" id="pwd" name="password">
14
15     <!-- Radio buttons -->
16     <p>Gender:</p>
17     <input type="radio" id="male" name="gender" value="male">
18     <label for="male">Male</label>
19     <input type="radio" id="female" name="gender" value="female">
20     <label for="female">Female</label>
21
22     <!-- Checkbox -->
23     <input type="checkbox" id="agree" name="agree">
24     <label for="agree">I agree to the Terms and Conditions</label>
25     >
26
27     <!-- Dropdown select -->
28     <label for="country">Country:</label>
29     <select id="country" name="country">
30         <option value="">-- Select Country --</option>
31         <option value="np">Nepal</option>
32         <option value="in">India</option>
33         <option value="us">United States</option>
34         <option value="uk">United Kingdom</option>
35     </select>

```

```

35
36 <!-- Textarea -->
37 <label for="message">Your Message:</label>
38 <textarea id="message" name="message" rows="5" cols="40"
39     placeholder="Write your message here..."></textarea>
40
41 <!-- Submit button -->
42 <button type="submit">Send Message</button>
43 <button type="reset">Clear Form</button>
44
45 </form>

```

Listing 15: A complete HTML form

10.1 Form Elements Explained

10.1.1 <form>

The container for all form controls. Key attributes:

- **action** — URL where form data is sent when submitted.
- **method** — HTTP method: **"get"** (data in URL) or **"post"** (data in body).

10.1.2 <input>

The most versatile form element. The **type** attribute controls what kind of input is shown:

type	value	Description
text		Single-line text field
email		Email address (validates format)
password		Masked text for passwords
number		Numeric input with optional min/max
checkbox		A tick-box (on/off)
radio		One choice from a group
file		File upload button
date		Date picker
submit		Button to submit the form
reset		Button to reset all fields
hidden		Hidden data sent with the form

10.1.3 <label>

The **<label>** tag creates a text label for a form control. Link it to an input using the **for** attribute (matching the input's **id**). Clicking the label focuses the associated input, which improves usability.

10.1.4 <textarea>

A multi-line text input. Use **rows** and **cols** to set the visible size.

10.1.5 <select> and <option>

Creates a **dropdown menu**. Each choice is an `<option>` element with a `value` attribute. Add `multiple` to allow selecting more than one option.

10.1.6 <button>

Creates a clickable button. The `type` attribute can be:

- `"submit"` — submits the form (default behaviour).
- `"reset"` — clears all form fields.
- `"button"` — no default action (used with JavaScript).

Tip

Always use `<label>` with every form control — it is essential for **accessibility**. Users with screen readers depend on labels to understand what each input is for.

11 Semantic HTML Tags

Semantic HTML tags describe the *meaning* or *role* of the content they contain, rather than just its appearance. HTML5 introduced many semantic elements to replace the overuse of `<div>` tags.

11.1 Why Use Semantic Tags?

- **Accessibility:** Screen readers understand the structure of the page.
- **SEO:** Search engines better understand page content and rank it higher.
- **Readability:** Code is easier for developers to read and maintain.
- **Consistency:** Provides a predictable structure across web projects.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>My Blog</title>
6 </head>
7 <body>
8
9   <!-- Site-wide header / logo / branding -->
10  <header>
11    <h1>My Personal Blog</h1>
12    <p>Thoughts, ideas, and discoveries.</p>
13  </header>
14
15  <!-- Navigation menu -->
16  <nav>
17    <a href="index.html">Home</a>
18    <a href="about.html">About</a>
```

```
19     <a href="contact.html">Contact</a>
20 </nav>
21
22 <!-- Main content area of the page -->
23 <main>
24
25     <!-- A standalone section of related content -->
26     <section>
27         <h2>Latest Posts</h2>
28
29         <!-- A self-contained piece of content (blog post, news
30              item) -->
31         <article>
32             <h3>Getting Started with HTML</h3>
33             <p>Posted on <time datetime="2024-09-01">1 September
34                 2024</time></p>
35             <p>HTML is the foundation of every web page. In this
36                 post,
37                 we explore the basics...</p>
38         </article>
39
40         <article>
41             <h3>Understanding CSS Flexbox</h3>
42             <p>Posted on <time datetime="2024-09-10">10 September
43                 2024</time></p>
44             <p>Flexbox makes it easy to create responsive layouts
45                 ...</p>
46         </article>
47
48     </section>
49
50     <!-- Supplementary content related to the main content -->
51     <aside>
52         <h3>About the Author</h3>
53         <p>A web developer sharing knowledge with the world.</p>
54     </aside>
55
56 </main>
57
58 <!-- Site-wide footer -->
59 <footer>
60     <p>Copyright 2024 My Blog. All rights reserved.</p>
61     <a href="privacy.html">Privacy Policy</a>
62 </footer>
63
64 </body>
65 </html>
```

Listing 16: Page layout using semantic tags

11.2 Overview of Semantic Elements

Tag	Purpose
<code><header></code>	Introductory content: logo, site title, main navigation
<code><nav></code>	A block of navigation links
<code><main></code>	The dominant content of the page (only one per page)
<code><section></code>	A thematic grouping of content with a heading
<code><article></code>	Self-contained content that could stand alone (blog post, news)
<code><aside></code>	Sidebar content — related but not central to the main content
<code><footer></code>	Footer of the page or a section: copyright, links, contact
<code><figure></code>	Self-contained media (image, diagram, code snippet)
<code><figcaption></code>	Caption for a <code><figure></code> element
<code><time></code>	Represents a date or time

Note

Before semantic tags were introduced, developers used `<div id="header">`, `<div id="nav">`, etc. Semantic tags replace these with purpose-built elements that are more meaningful and accessible.

12 HTML Attributes

Attributes provide additional information about HTML elements. They are placed inside the **opening tag** and follow the format:

```
1 <tagname attribute="value">Content</tagname>
```

Multiple attributes can be added to one tag, separated by spaces:

```
1 <input type="text" id="username" class="form-input" placeholder="Enter name">
```

12.1 id

The `id` attribute gives an element a **unique identifier** on the page. No two elements should share the same `id`.

Uses:

- CSS styling: `#myId { color: red; }`
- JavaScript selection: `document.getElementById("myId")`
- In-page links: `<a href="#myId">Jump</a>`

```
1 <h2 id="introduction">Introduction</h2>
2 <a href="#introduction">Go to Introduction</a>
```

12.2 class

The **class** attribute assigns one or more **reusable class names** to an element. Multiple elements can share the same class. Multiple classes are separated by spaces.

```
1 <p class="highlight">This paragraph is highlighted.</p>
2 <p class="highlight bold">This one is highlighted and bold.</p>
3 <button class="btn btn-primary">Click Me</button>
```

12.3 href

The **href** (Hypertext Reference) attribute specifies the **destination URL** for **<a>** tags and the location of stylesheets for **<link>** tags.

```
1 <a href="https://www.wikipedia.org">Visit Wikipedia</a>
2 <link rel="stylesheet" href="styles.css">
```

12.4 src

The **src** (source) attribute specifies the **path to a file** to embed — used with ****, **<script>**, and **<iframe>**.

```
1 
2 <script src="app.js"></script>
```

12.5 alt

The **alt** attribute provides **alternative text** for images. It is displayed when the image cannot load and is read by screen readers.

```
1 
```

12.6 target

Used with **<a>** tags to specify **where to open** the linked page:

Value	Behaviour
<code>_self</code>	Opens in the same tab (default)
<code>_blank</code>	Opens in a new tab or window
<code>_parent</code>	Opens in the parent frame
<code>_top</code>	Opens in the full browser window

```
1 <a href="https://www.example.com" target="_blank"
2   rel="noopener noreferrer">Open in New Tab</a>
```

13 Putting It All Together — A Complete Example

Here is a complete HTML page that uses all the concepts covered in these notes:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-
6     scale=1.0">
7   <title>HTML Demo Page</title>
8 </head>
9 <body>
10
11   <header>
12     <h1>HTML5 Demo Page</h1>
13     <nav>
14       <a href="#about">About</a> |
15       <a href="#skills">Skills</a> |
16       <a href="#contact">Contact</a>
17     </nav>
18   </header>
19
20   <hr>
21
22   <main>
23
24     <section id="about">
25       <h2>About HTML</h2>
26       <p>
27         HTML stands for <strong>HyperText Markup Language</
28         strong>. It is the
29         <em>standard language</em> for creating web pages. The
30         current version
31         is <mark>HTML5</mark>.
32       </p>
33       
34     </section>
35
36     <section id="skills">
37       <h2>What You Will Learn</h2>
38
39       <h3>Topics Covered</h3>
40       <ul>
41         <li>Document Structure</li>
42         <li>Text Formatting</li>
43         <li>Links and Images</li>
44         <li>Lists, Tables, and Forms</li>
45         <li>Semantic HTML</li>
46       </ul>
47
48       <h3>Learning Roadmap</h3>
```

```

46     <ol>
47         <li>Learn HTML basics</li>
48         <li>Style pages with CSS</li>
49         <li>Add interactivity with JavaScript</li>
50         <li>Build real projects</li>
51     </ol>
52
53     <h3>Comparison Table</h3>
54     <table border="1">
55         <thead>
56             <tr>
57                 <th>Technology</th>
58                 <th>Role</th>
59                 <th>Analogy</th>
60             </tr>
61         </thead>
62         <tbody>
63             <tr>
64                 <td>HTML</td>
65                 <td>Structure</td>
66                 <td>Skeleton of a building</td>
67             </tr>
68             <tr>
69                 <td>CSS</td>
70                 <td>Styling</td>
71                 <td>Paint and decoration</td>
72             </tr>
73             <tr>
74                 <td>JavaScript</td>
75                 <td>Behaviour</td>
76                 <td>Electricity and plumbing</td>
77             </tr>
78         </tbody>
79     </table>
80 </section>
81
82 <section id="contact">
83     <h2>Contact Us</h2>
84     <form action="/submit" method="post">
85
86         <label for="name">Name:</label><br>
87         <input type="text" id="name" name="name"
88             placeholder="Your full name"><br><br>
89
90         <label for="email">Email:</label><br>
91         <input type="email" id="email" name="email"
92             placeholder="you@example.com"><br><br>
93
94         <label for="msg">Message:</label><br>
95         <textarea id="msg" name="message" rows="4"
96             cols="40"></textarea><br><br>

```

```
97
98     <label for="topic">Topic:</label>
99     <select id="topic" name="topic">
100         <option value="general">General Enquiry</option>
101         <option value="support">Technical Support</option>
102         <option value="feedback">Feedback</option>
103     </select><br><br>
104
105     <button type="submit">Send Message</button>
106 </form>
107 </section>
108
109 </main>
110
111 <hr>
112
113 <footer>
114     <small>Copyright 2024 HTML Demo. All rights reserved.</
115         small>
116 </footer>
117 </body>
118 </html>
```

Listing 17: A complete HTML5 web page

14 Quick Reference Cheat Sheet

Tag / Attribute	Description
<code><!DOCTYPE html></code>	Declares HTML5 document
<code><html></code>	Root element of the page
<code><head></code>	Container for metadata
<code><title></code>	Browser tab title
<code><body></code>	All visible page content
<code><h1>- <h6></code>	Headings, largest to smallest
<code><p></code>	Paragraph
<code>
</code>	Line break (self-closing)
<code><hr></code>	Horizontal rule (self-closing)
<code>/ </code>	Bold / Strongly important
<code><i>/ </code>	Italic / Emphasised
<code><u></code>	Underlined text
<code><mark></code>	Highlighted text
<code><small></code>	Smaller text
<code></code>	Hyperlink
<code></code>	Image
<code>/ </code>	Unordered / Ordered list
<code></code>	List item
<code><table></code>	Table container
<code><tr></code>	Table row
<code><th></code>	Table header cell
<code><td></code>	Table data cell
<code><form></code>	Form container
<code><input></code>	Input field (self-closing)
<code><label></code>	Label for an input
<code><textarea></code>	Multi-line text input
<code><select></code>	Dropdown menu
<code><button></code>	Clickable button
<code><header></code>	Page or section header
<code><nav></code>	Navigation links
<code><main></code>	Main content area
<code><section></code>	Thematic section
<code><article></code>	Self-contained content
<code><aside></code>	Sidebar / supplementary content
<code><footer></code>	Page or section footer
<code>id="..."</code>	Unique identifier for an element
<code>class="..."</code>	Reusable class name(s)
<code>href="..."</code>	Link destination URL
<code>src="..."</code>	Source file path or URL
<code>alt="..."</code>	Alternative text for images
<code>target="..."</code>	Where to open a link

Summary and Next Steps

Congratulations on completing these HTML fundamentals! Here is a summary of what you have learned:

- **HTML** is the backbone of every web page, using tags to mark up content.
- Every page follows a standard structure: `!DOCTYPE`, `html`, `head`, `title`, and `body`.
- **Headings** (`h1–h6`), **paragraphs**, **line breaks**, and **horizontal rules** organise and format text.
- **Text formatting tags** add visual and semantic meaning.
- **Links** and **images** connect and enrich page content.
- **Lists** and **tables** organise structured data.
- **Forms** collect user input through various input controls.
- **Semantic tags** improve accessibility, SEO, and code readability.
- **Attributes** like `id`, `class`, `href`, `src`, `alt`, and `target` control element behaviour and identity.

Recommended Next Steps

1. **Practice:** Build 3–5 simple HTML pages from scratch (a personal profile, a recipe page, a contact form).
2. **Learn CSS:** Style your HTML pages with colours, fonts, spacing, and layouts.
3. **Try Flexbox and Grid:** Modern CSS layout techniques for responsive design.
4. **Learn JavaScript:** Add interactivity — button clicks, form validation, dynamic content.
5. **Use a code editor:** VS Code with the Live Server extension lets you see changes instantly in your browser.
6. **Validate your HTML:** Use the W3C Markup Validator at <https://validator.w3.org> to check your code for errors.

Tip

The best way to learn HTML is by **doing**. Open a text editor, type some code, save it as `index.html`, and open it in your browser. Experiment — you cannot break anything!

End of Web Fundamentals & HTML Basics — Notes
Happy coding!
